

CROWN INSTITUTE

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FROM POSSIBILITY TO ACTUALITY

Big Data (ICT303)

LECTURE 8

Big Data Tools: Hadoop, YARN, Pig

Introduction

- The 5 V’s of Big Data
- Hadoop
- SQL on Hadoop
- Apache Spark

The 5 V's of Big Data

- Every minute:
  - More than 300000 tweets are created
  - Netflix subscribers are streaming more than 70000 hours of video at once
  - Apple users download 30000 apps
  - Instagram users like almost 2 million photos
- Big Data encompasses both structured and highly unstructured forms of data

The 5 V's of Big Data

- **Volume**: the amount of data, also referred to the data "at rest"
- **Velocity**: the speed at which data comes in and goes out, data "in motion"
- **Variety**: the range of data types and sources that are used, data in its "many forms"
- **Veracity**: the uncertainty of the data, data "in doubt"
- **Value**: TCO and ROI of the data

The 5 V's of Big Data

- Examples:
  - Large scale enterprise systems (e.g., ERP, CRM, SCM)
  - Social networks (e.g., Twitter, Weibo, WeChat)
  - Internet of Things
  - Open data

Hadoop

- Open-source software framework used for distributed storage and processing of big datasets
- Can be set up over a cluster of computers built from normal, commodity hardware
- Many vendors offer their implementation of a Hadoop stack (e.g. Amazon, Cloudera, Dell, Oracle, IBM, Microsoft)

Hadoop

- History of Hadoop
- The Hadoop stack

History of Hadoop

- Key building blocks:
  - Google File System: a file system that could be easily distributed across commodity hardware, whilst providing fault tolerance
  - Google MapReduce: a programming paradigm to write programs that can be automatically parallelized and executed across a cluster of different computers
- Nutch web crawler prototype developed by Doug Cutting
  - Later renamed to Hadoop
- In 2008, Yahoo! open-sourced Hadoop as “Apache Hadoop”

The Hadoop Stack

- Four modules:
  - Hadoop Common: a set of shared programming libraries used by the other modules
  - Hadoop Distributed File System (HDFS): a Java-based file system to store data across multiple machines
  - MapReduce framework: a programming model to process large sets of data in parallel
  - YARN (Yet Another Resource Negotiator): handles the management and scheduling of resource requests in a distributed environment

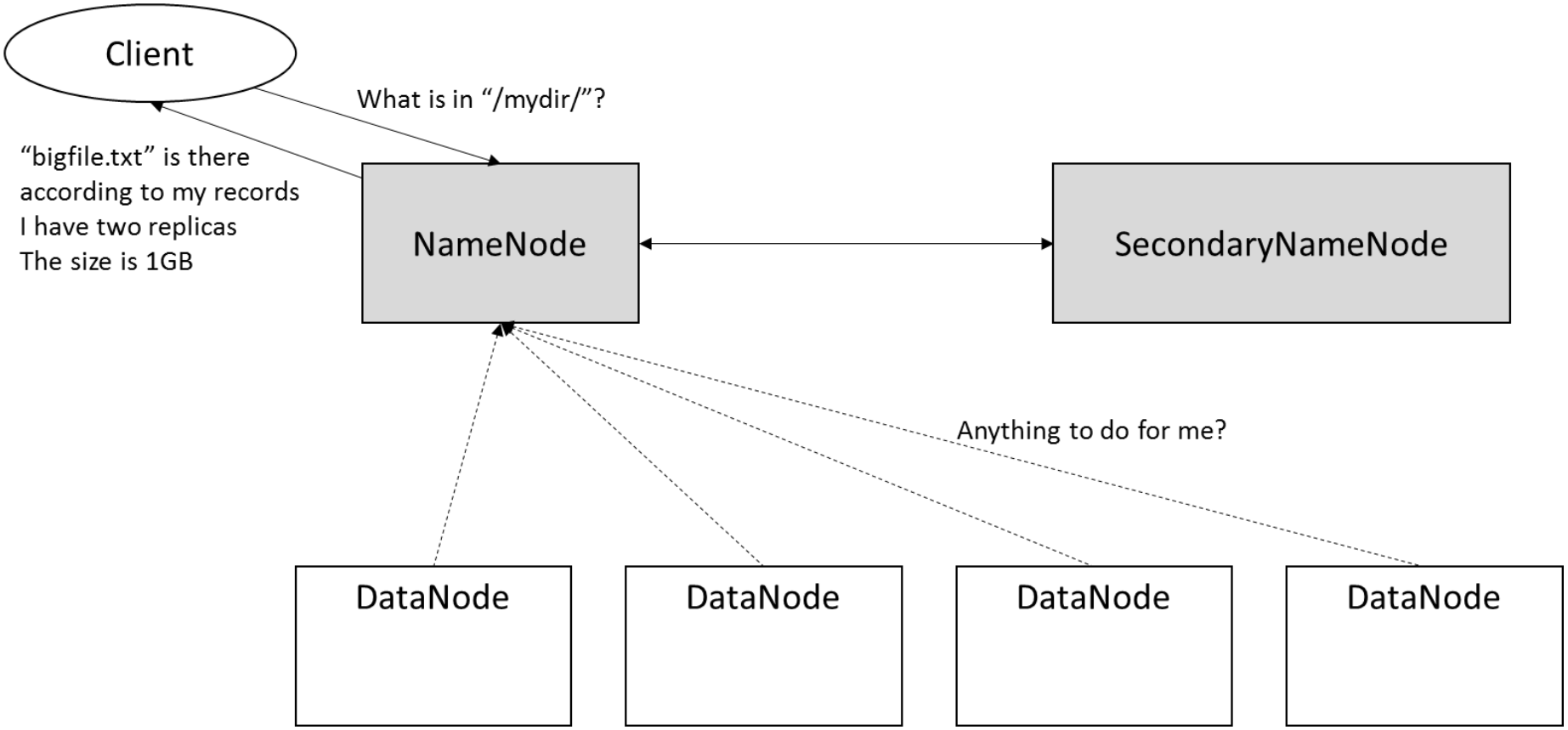
Hadoop Distributed File System (HDFS)

- Distributed file system to store data across a cluster of commodity machines
- High emphasis on fault-tolerance
- HDFS cluster is composed of a NameNode and various DataNodes

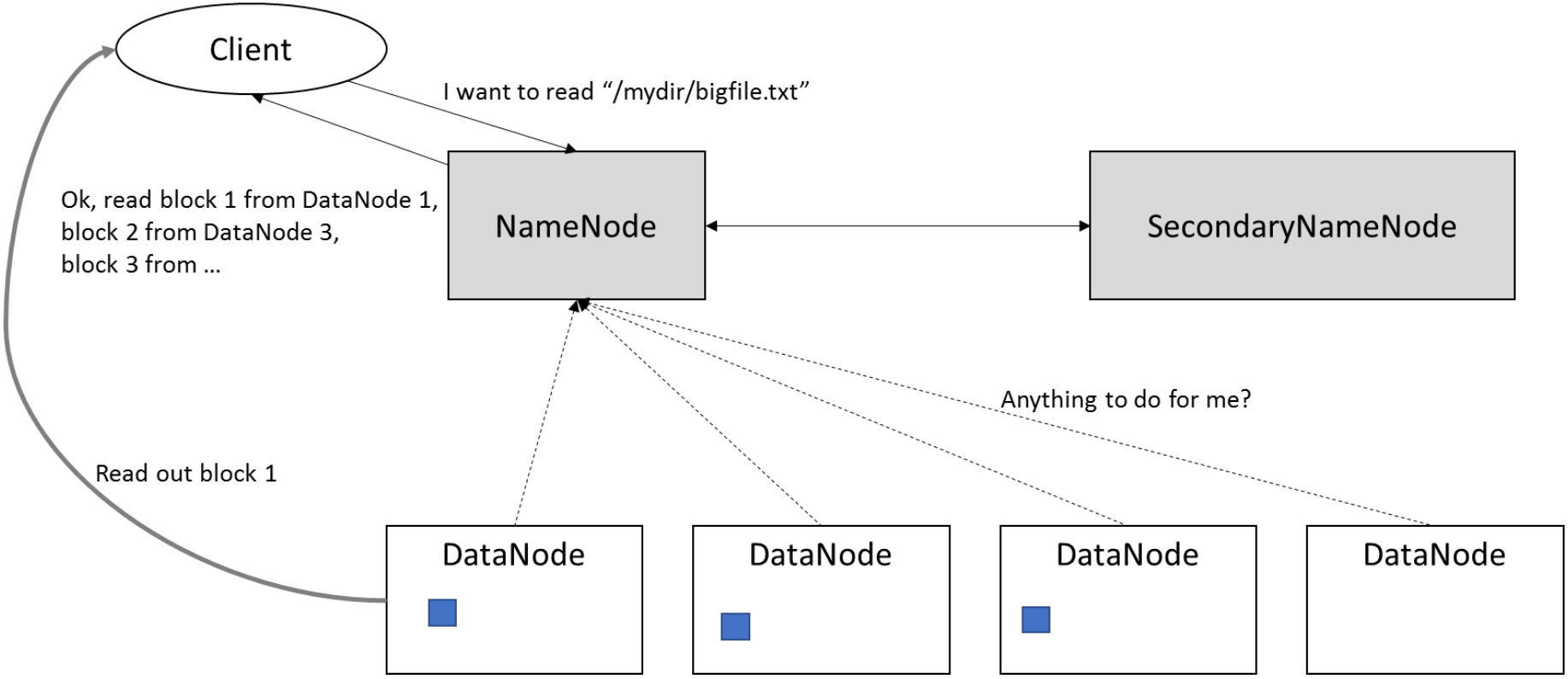
Hadoop Distributed File System (HDFS)

- NameNode
  - a server which holds all the metadata regarding the stored files
  - manages incoming file system operations
  - maps data blocks (parts of files) to DataNodes
- DataNode
  - handles file read and write requests
  - create, delete and replicate data blocks amongst their disk drives
  - continuously loop, asking the NameNode for instructions.
- Note: size of 1 data block is typically 64 megabytes

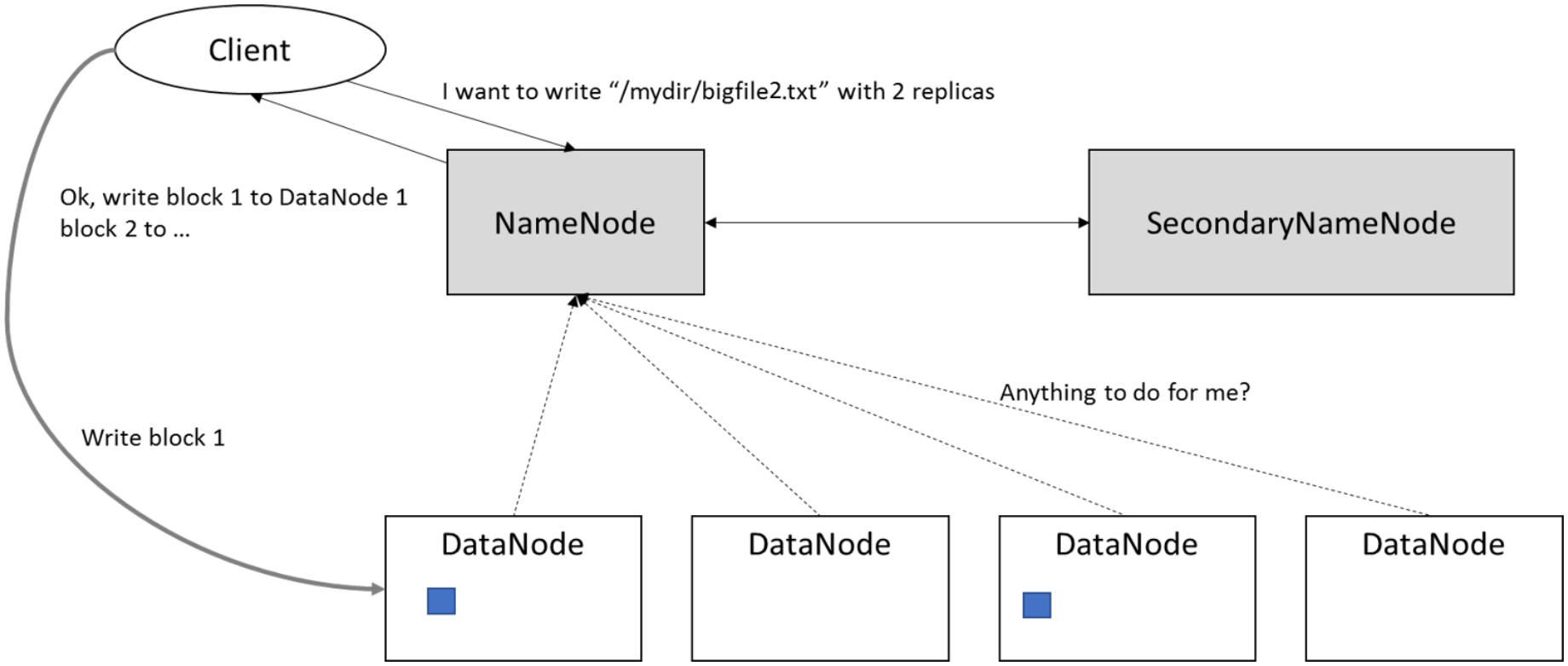
Hadoop Distributed File System (HDFS)



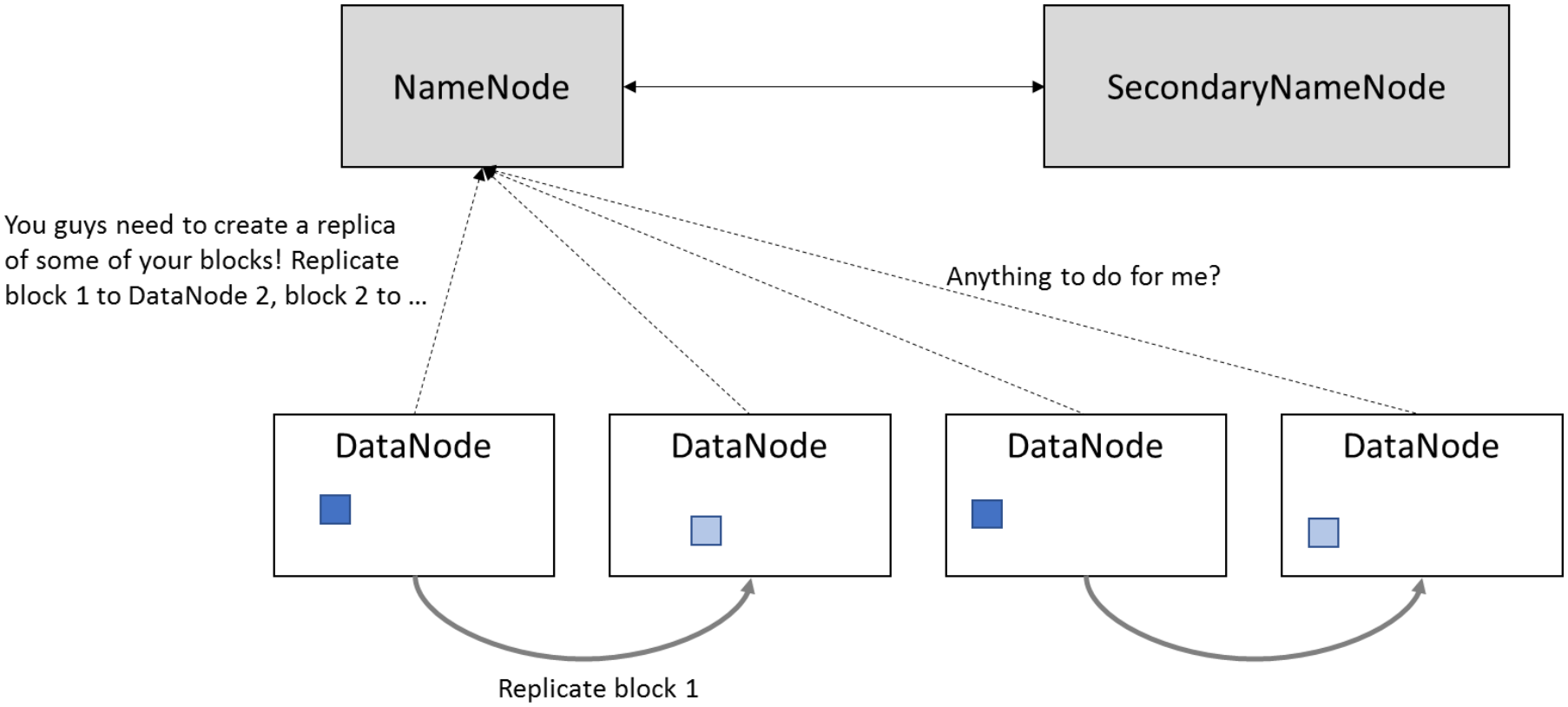
Hadoop Distributed File System (HDFS)



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Hadoop Distributed File System (HDFS)



Hadoop Distributed File System (HDFS)

- HDFS provides a native Java API to allow for writing Java programs that can interface with HDFS

```
String filePath = “/data/all_my_customers.csv”;
Configuration config = new Configuration();
org.apache.hadoop.fs.FileSystem hdfs =
org.apache.hadoop.fs.FileSystem.get(config);
org.apache.hadoop.fs.Path path = new
org.apache.hadoop.fs.Path(filePath);
org.apache.hadoop.fs.FSDataInputStream inputStream =
hdfs.open(path);
byte[] received = new byte[inputStream.available()];
inputStream.readFully(received);
```

Hadoop Distributed File System (HDFS)

```
// ...
org.apache.hadoop.fs.FSDataInputStream inputStream = hdfs.open(path);
byte[] buffer=new byte[1024]; // Only handle 1KB at once
int bytesRead;
while ((bytesRead = in.read(buffer)) > 0) {
    // Do something with the buffered block here
}
```

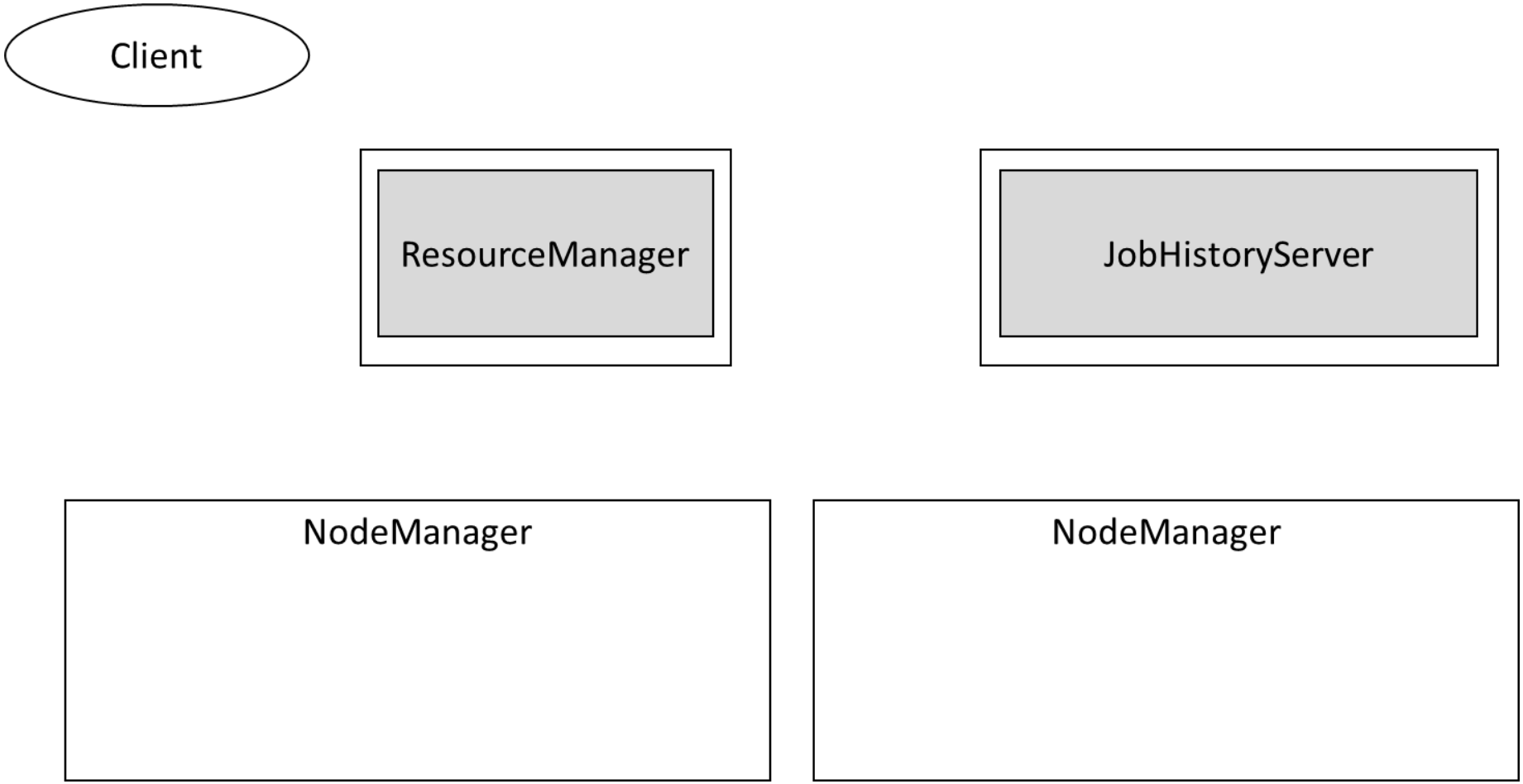
Hadoop Distributed File System (HDFS)

`hadoop fs -mkdir mydir`	Create a directory on HDFS
`hadoop fs -ls`	List files and directories on HDFS
`hadoop fs -cat myfile`	View a file's content
`hadoop fs -du`	Check disk space usage on HDFS
`hadoop fs -expunge`	Empty trash on HDFS
`hadoop fs -chgrp mygroup myfile`	Change group membership of a file on HDFS
`hadoop fs -chown myuser myfile`	Change file ownership of a file on HDFS
`hadoop fs -rm myfile`	Delete a file on HDFS
`hadoop fs -touchz myfile`	Create an empty file on HDFS
`hadoop fs -stat myfile`	Check the status of a file (file size, owner, …)
`hadoop fs -test -e myfile`	Check if file exists on HDFS
`hadoop fs -test -z myfile`	Check if file is empty on HDFS
`hadoop fs -test -d myfile`	Check if myfile is a directory on HDFS

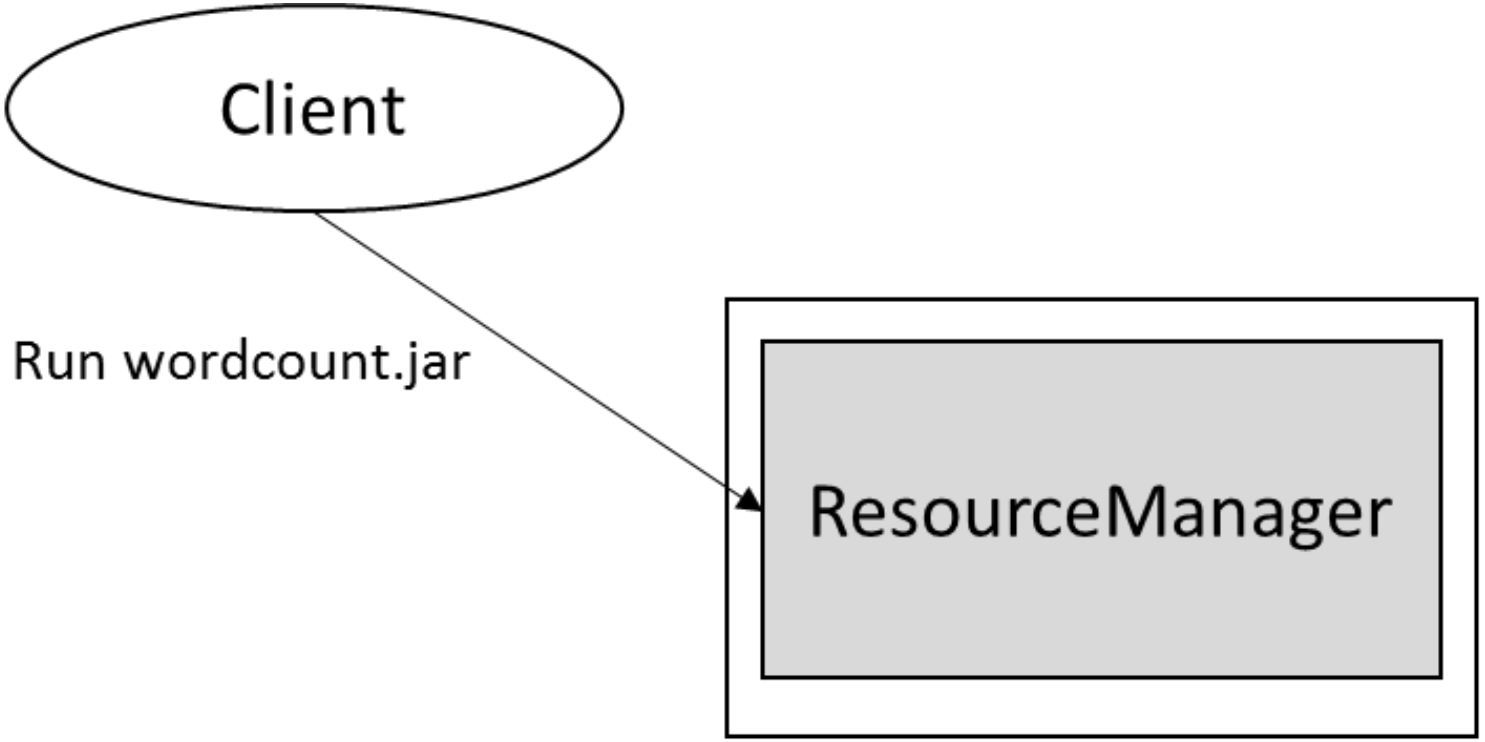
Yet Another Resource Negotiator (YARN)

- Yet Another Resource Negotiator (YARN) distributes a MapReduce program across different nodes and takes care of coordination
- Three important services
  - ResourceManager: a global YARN service that receives and runs applications (e.g., a MapReduce job) on the cluster
  - JobHistoryServer: keeps a log of all finished jobs
  - NodeManager: responsible to oversee resource consumption on a node

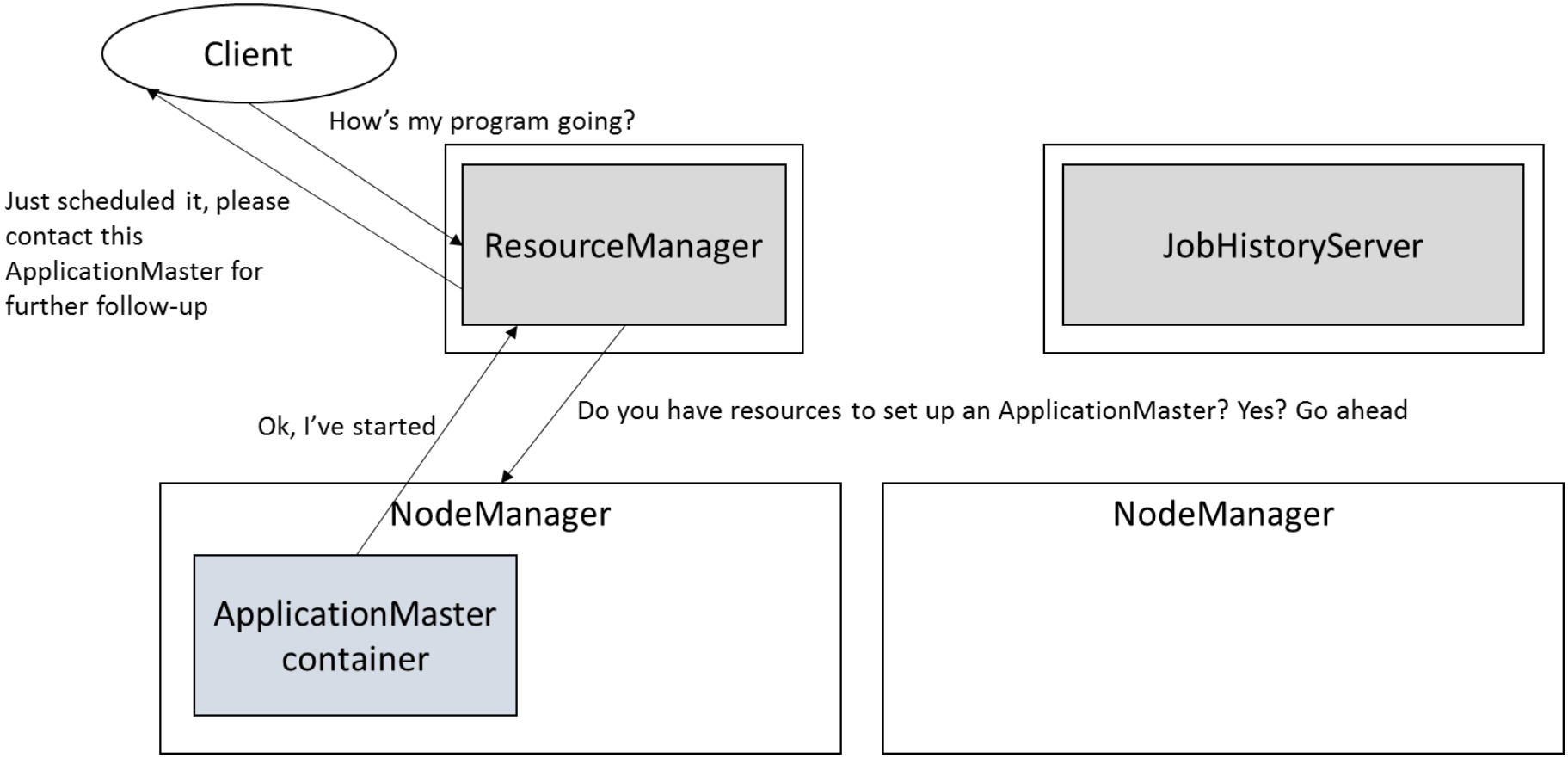
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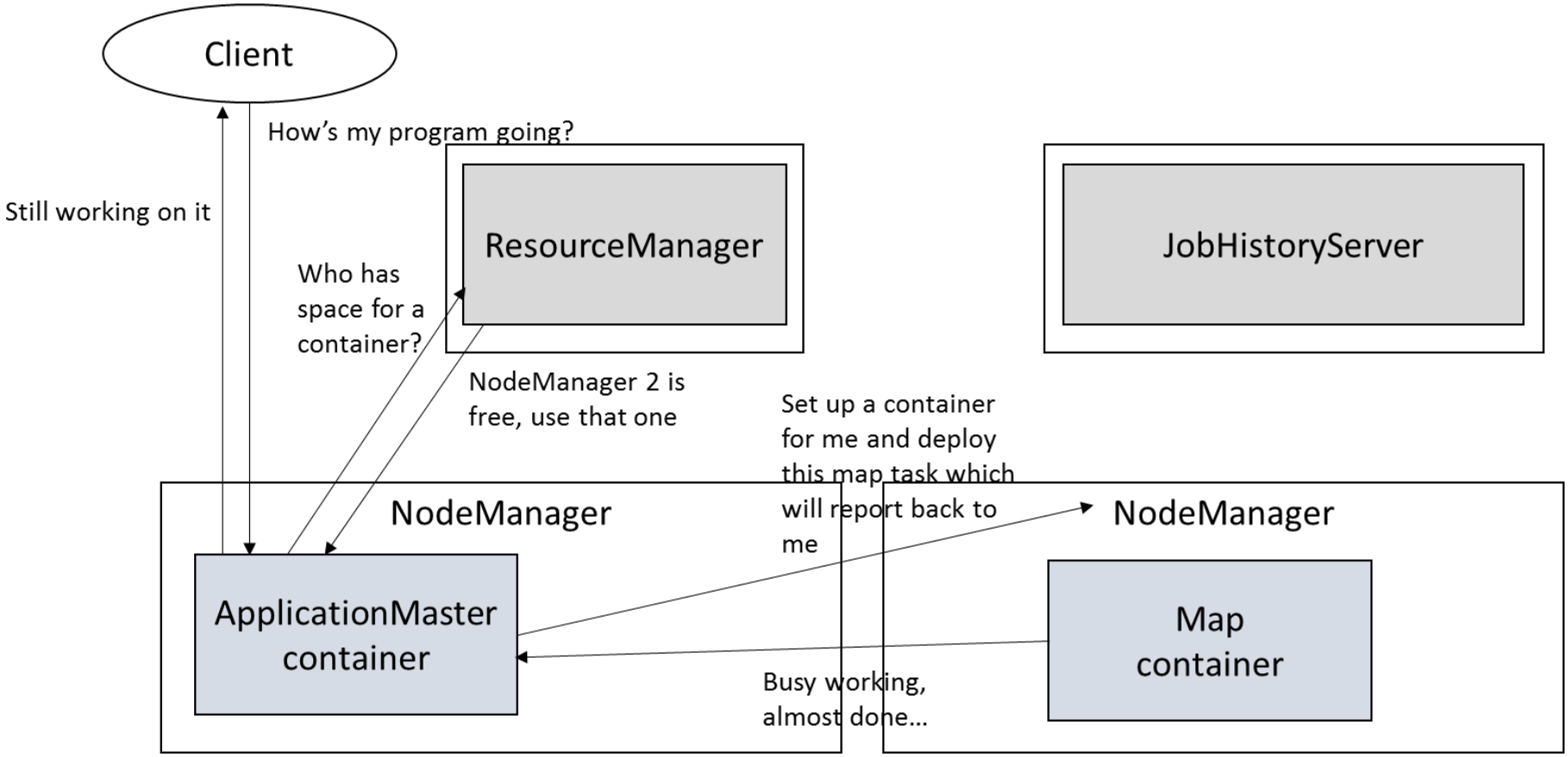
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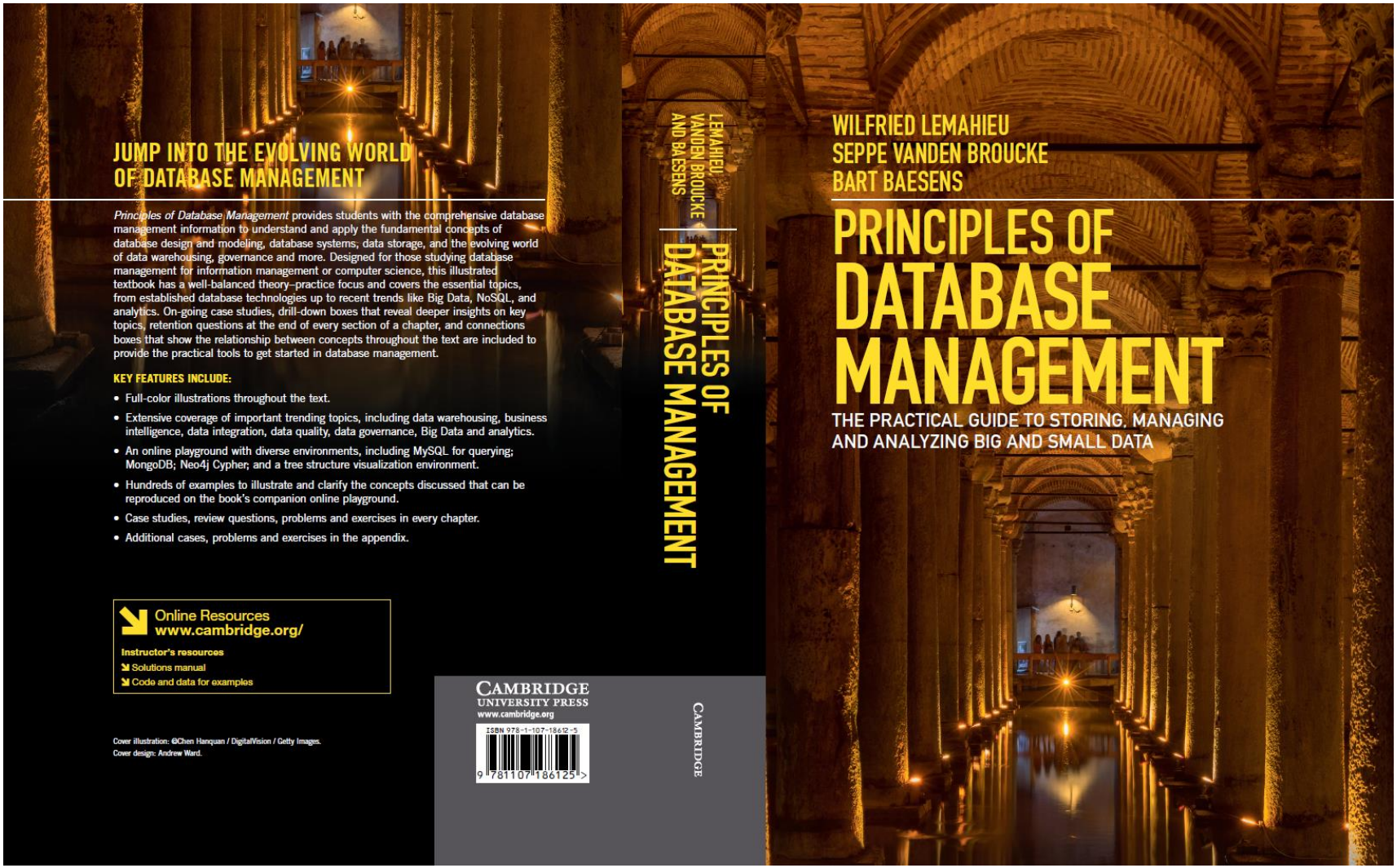
Yet Another Resource Negotiator (YARN)



Yet Another Resource Negotiator (YARN)

- Complex setup
- Allows to run programs and applications other than MapReduce

Source



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